DOI: 10.1111/j.1464-5491.2009.02732.x

Original Article: Care Delivery

A Ubiquitous Chronic Disease Care system using cellular phones and the internet

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Accepted 4 April 2009

Abstract

Aims The rapidly increasing prevalence of chronic diseases is an important challenge to healthcare systems worldwide. To improve the quality and efficiency of chronic disease care, we investigated the effectiveness and applicability of the Ubiquitous Chronic Disease Care (UCDC) system using cellular phones and the internet for overweight patients with both Type 2 diabetes and hypertension.

Methods We conducted a randomized, controlled clinical trial over 3 months that included 123 patients at a university hospital and a community public health centre.

Results After 12 weeks, there were significant improvements in HbA_{1c} in the intervention group (7.6 ± 0.9% to 7.1 ± 0.8%, $P < 0.001$) compared with the control group (7.4 ± 0.9% to 7.6 ± 1.0%, $P = 0.03$). Furthermore, we observed a significant reduction in systolic and diastolic blood pressure, as well as improvements in total cholesterol, low-density lipoprotein-cholesterol and triglyceride levels in the intervention group. Furthermore, there was a significant increase in adiponectin levels in the intervention group compared with the control group, although high-sensitivity C-reactive protein and interleukin-6 levels did not change in either group.

Conclusions The novel UCDC system presented in this paper improved multiple metabolic parameters simultaneously in overweight patients with both Type 2 diabetes and hypertension.

Diabet. Med. 26, 628–635 (2009)

Keywords chronic disease, diabetes, hypertension, internet, obesity

Abbreviations ADA, American Diabetes Association; baPWV, brachial-ankle pulse wave velocity; BMI, body mass index; CRP, C-reactive protein; CV, coefficient of variation; HbA_{1c}, glycated haemoglobin; HBPM, home blood pressure measurement; hsCRP, high-sensitivity C-reactive protein; IL-6, interleukin-6; LDL, low-density lipoprotein; OBPM, office blood pressure measurement; SMBG, self-monitoring blood glucose; SMS, short message service; UCDC, Ubiquitous Chronic Disease Care

Introduction

The rapidly increasing prevalence of chronic diseases such as Type 2 diabetes, obesity, hypertension and dyslipidaemia is an important challenge to healthcare systems worldwide. There is an urgent need to reform health services from provider-focused episodic care to a more patient-centred continuous care model [1]. The present healthcare system has a limited capacity for facilitating self-management of disease, including life-style interventions, over the course of a lifetime. A new system for the management of chronic diseases that allows patients to control their healthcare in real-life situations is required.

Widespread internet access makes the internet a viable and logical mode for delivering health interventions. Households with personal computers grew from 8% in 1984 to > 65% by 2008 in the UK [2,3]. Various clinical trials have attempted to utilize the internet as a novel healthcare delivery device. Several

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studies performed in Korea have investigated the effects of internet-based monitoring of blood glucose control in patients with Type 2 diabetes [4–6]. McKay *et al.* [7] reported that an internet-based supplement to usual care for sedentary Type 2 diabetic patients enhanced physical activity and Tate *et al.* [8] showed that internet-based behaviour counselling significantly improved weight loss in overweight and obese adults. Likewise, the widespread distribution of cellular phones, combined with their unique ability to communicate data in real time, makes them an ideal novel heathcare delivery tool for chronic disease. Recently, Quinn *et al.* [9] demonstrated that a mobile diabetes management programme improved blood glucose control in Type 2 diabetes and Coberley *et al.* [10] reported that telephone calls increased the rates of testing for low-density lipoprotein (LDL)-cholesterol and glycated haemoglobin (HbA_{1c}) in previously non-adherent diabetic patients.

However, multifactorial interventions targeting blood glucose, blood pressure, and weight control simultaneously have not been reported in the field of e-medicine. Multifactorial intervention aimed at treating these risk factors in patients with Type 2 diabetes could reduce the risk of cardiovascular and microvascular events by approximately 50% [11]. This could eventually reduce mortality rates from many causes, including cardiovascular disease, as estimated by the Steno-2 study [12]. Therefore, in the present study we simultaneously controlled multiple metabolic parameters of overweight patients with both Type 2 diabetes and hypertension using e-medicine. Moreover, our novel e-medicine system took advantage of both cellular phones and the internet. Until now, most studies involving e-medicine have used either a website or a mobile phone, but not both [4,5,7,8,13,14]. Some concerns have been raised about the ease of use of websites in internet-based studies, whereas using cellular phones alone leads to lack of systematic care.

Our goal was to examine whether a Ubiquitous Chronic Disease Care (UCDC) system using both the internet and cellular phones could facilitate chronic disease self-management and improve multiple metabolic parameters in overweight patients with both Type 2 diabetes and hypertension better than conventional healthcare in out-patient clinics. We determined the effect of the UCDC system on levels of adiponectin and proinflammatory markers such as C-reactive protein (CRP) and interleukin (IL)-6, as well as arterial stiffness, which are known cardiovascular risk factors.

Patients and methods

Study participants

We recruited patients for this open-label, randomized, controlled, prospective study from both a university hospital setting (Korea University) and a community healthcare centre (Guro-Gu Public Health Centre; Fig. 1). We enrolled 123 subjects, between 30 and 70 years of age, who met the following criteria: (i) a diagnosis of both Type 2 diabetes and hypertension at least 1 year previously by a physician; (ii) HbA_{1c} 6.5–10.0%; (iii) blood pressure > 130/80 mmHg; and (iv) body mass index (BMI) ≥ 23.0 kg/m^2 (overweight according to Asia-Pacific criteria) [15]. Our exclusion criteria included (i) severe diabetic complications (e.g. diabetic foot or severe diabetic retinopathy); (ii) liver dysfunction with aspartate aminotransferase or alanine aminotransferase > 2.5 times the reference level, or renal dysfunction (serum creatinine > 132 μmol/l); (iii) medical history of congestive heart failure, angina pectoris, myocardial infarction, or stroke based on a physician's diagnosis; (iv) pregnancy or lactation; or (v) other medical problems that could affect study results or trial participation. All participants provided written informed consent. The Korea University Institutional Review Board, in accordance with the Declaration of Helsinki of the World Medical Association, approved this study protocol.

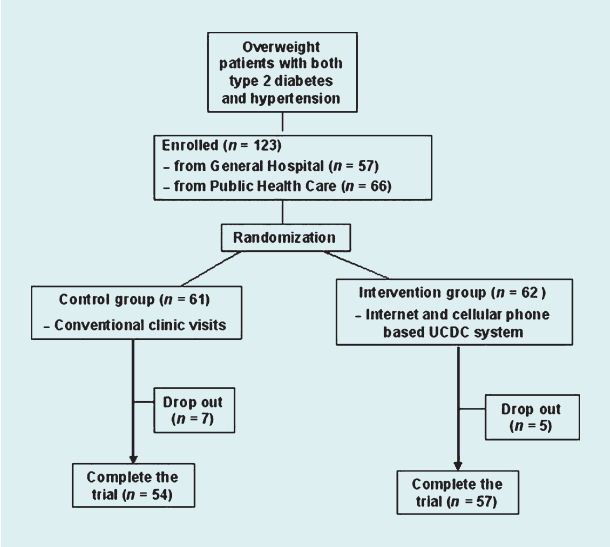

FIGURE 1 Flow chart of randomized controlled intervention trial.

The UCDC system

An outline of the UCDC system used in this study is summarized in Fig. 2. Patients in the intervention groups received a cellular phone (LG-SV280; LG Electronics, Seoul, Korea) with a modular blood glucose measuring device (Anycheck; Insung Information Co., Seoul, Korea), strips, and lancets. They also received an automatic blood pressure monitoring device (T5M; Omron, Kyoto, Japan), as well as body weight scales (HD308; Tanita, Tokyo, Japan).

First, the UCDC system sent out an alarm on the cellular phone to remind the participant to measure their blood glucose, blood pressure twice a day (before breakfast and bedtime) and body weight once a day (before breakfast). The Anycheck device attached to their cellular phone conducted the glucose measurements and automatically sent the results to a central study database. As soon as participants transmitted their glucose

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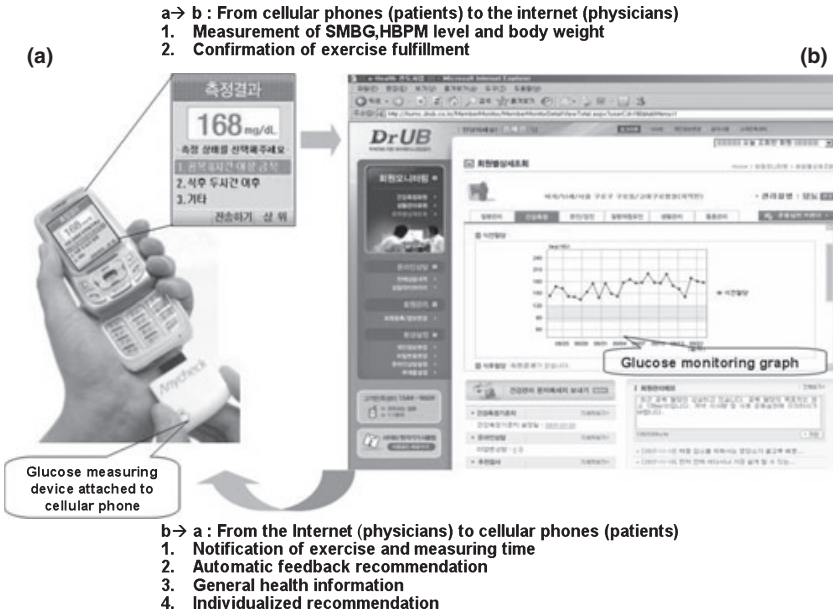

FIGURE 2 Outline of the Ubiquitous Chronic Disease Care (UCDC) system composed of a patient-based cellular phone (a) and web-based physician communication (b). HBPM, home blood pressure measurement; SMBG, self-monitoring blood glucose.

measurement through their cellular phones, they immediately received messages of encouragement, reminders, and recommendations according to a pre-defined algorithm that was developed by endocrinologists, dieticians and nurses at Korea University based on the American Diabetes Association (ADA) Guidelines [16] and the Korean Staged Diabetes Management Guidelines [17]. Examples of reminder messages are:

- Your fasting blood glucose level is very high compared with the appropriate target level for Type 2 diabetes (< 7.2 mmol/l). If this high level recurs often, diabetic complications might result. Reduce your calorie intake and avoid foods high in fat. In addition, plan for regular exercise after your meals.
- Your systolic blood pressure level is currently at the prehypertensive stage. The appropriate target of systolic blood pressure for patients with Type 2 diabetes is below 130 mmHg. Reduce blood pressure by getting regular exercise and avoiding salty foods and alcohol.

Second, the UCDC system automatically recorded participant's exercise time using the short message service (SMS), which was predefined according to each patient's daily schedule. Participants replied via an automatic answer system whether or not they actually exercised. Exercise of at least 150 min/week, distributed over at least 3 days each week, was recommended based on the ADA Guidelines [16].

Third, participants received information via SMS three times a day regarding healthy diet and exercise methods, along with general information about diabetes, hypertension and obesity. Furthermore, using the internet website, physicians could follow participant's trends in blood glucose levels, blood pressure and body weight changes, allowing them to send individualized recommendations to patients when needed (http://kumc.drub.co.kr).

Patients in the control group visited their clinic according to their routine schedule and received the usual out-patient treatment from their physicians during the study period. During the trial, drug dosage was not changed in either the UCDC or the control groups at either location.

Anthropometric and laboratory variables

BMI was calculated as weight/height^2 (kg/m^2). Blood chemistry was analysed at the Korea University Guro Hospital laboratory (Seoul, Korea). The glucose oxidase method was employed to measure plasma glucose. A human insulin-specific radioimmunoassay kit (Linco Research Inc., St Charles, MO, USA) was used to measure insulin levels, with a coefficient of variation (CV) of 2.2%. This kit had a cross-reactivity of < 0.2% with human proinsulin. Insulin resistance was calculated by the homeostasis model assessment. Serum total cholesterol, triglycerides, and high-density lipoprotein-cholesterol were determined by enzymatic methods using a chemical analyser (Hitachi 747, Tokyo, Japan). HbA_{1c} was analysed by high-performance liquid chromatography using a Variant II analyser (Bio-Rad Laboratories, Hercules, CA, USA). Plasma adiponectin levels were measured using an adiponectin enzyme immunoassay kit (Phoenix Pharmaceuticals, Belmont, CA, USA), with a CV of 3.2%. Plasma IL-6 levels were measured using a Quantikine kit (R&D Systems, Belmont, CA, USA) with a CV of 8.1%. Plasma high-sensitivity (hs) CRP levels were measured using a hsCRP kit (Immunodiagno, Benshaim, Germany) with a CV of 9.2%. We excluded all subjects with hsCRP ≥ 15.0 mg/l to

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rule out any occult inflammatory or infectious disorders [18]. After a subject had rested in the supine position for 5 min, brachial-ankle pulse wave velocity (baPWV) was measured using a volume-plethysmographic apparatus (model BP-203RPE II; Colin, Komaki, Japan). The interobserver and intraobserver CVs were 8.4% and 10.0%, respectively.

Statistical analysis

Data were expressed as the mean ± SD, or median (interquartile range). We conducted an intention-to-treat analysis, and analysed non-normally distributed variables after log transformation. We used the Mann–Whitney *U*-test, Student's *t*-test and chi-squared test to examine the baseline differences between the control and intervention groups. Wilcoxon's signed rank test or paired-sample *t*-test was used to compare variables before and after intervention within each group. We performed repeated measures analysis of variance (ANOVA) to determine the interaction effect between measurement times and groups after adjusting for age, gender, current smoker status, duration of diabetes, and change of weight. In case of non-normally distributed variables, we transformed the original values to ranked ones prior to adopting the repeated-measures ANOVA. All statistical analyses based on two-sided test were performed using SPSS for Windows (Version 12.0; SPSS Inc., Chicago, IL, USA). Any $P < 0.05$ was regarded as statistically significant.

Results

Baseline characteristics of participants

The baseline characteristics of the 111 participants who finished the study are shown in Table 1. There were no significant differences in age, gender, duration of diabetes or hypertension, BMI, blood pressure, HbA_{1c}, insulin resistance, or lipid profiles between the two groups.

Changes in chronic disease control

After 12 weeks, a significant reduction was seen in body weight, BMI and waist circumference in both groups. However, we observed a significant reduction in the systolic and diastolic office blood pressure measurement (OBPM) after implementation of the UCDC system only in the intervention group. There was no improvement in OBPM in the control group (Table 2). Furthermore, the average home blood pressure measurement (HBPM) for the intervention

Table 1 Baseline characteristics of participants

	Control group (*n* = 54)	Intervention group (*n* = 57)	*P*-value
Age (years)	59.4 ± 8.4	57.0 ± 9.1	0.15
Sex (male %)	35 (64.8%)	30 (52.6%)	0.20*
Duration of diabetes (years)	7.2 ± 6.0	6.0 ± 5.4	0.25
Duration of hypertension (years)	4.2 ± 4.7	3.2 ± 3.5	0.23
Current smoker (yes: %)	17 (31.4%)	11 (19.2%)	0.11*
Weight (kg)	67.7 ± 10.8	66.4 ± 12.5	0.55
BMI (kg/m^2)	25.5 ± 3.3	25.6 ± 3.5	0.86
Waist circumference (cm)	91.3 ± 7.5	89.5 ± 9.7	0.28
Systolic blood pressure (mmHg)	138 ± 18	140 ± 18	0.47
Diastolic blood pressure (mmHg)	83 ± 10	84 ± 10	0.64
Right baPWV (cm/s)	1691 ± 377	1671 ± 382	0.78
Left baPWV (cm/s)	1695 ± 386	1648 ± 335	0.49
Fasting glucose (mmol/l)	7.4 ± 1.8	7.9 ± 1.8	0.27
HbA_{1c} (%)	7.4 ± 0.9	7.6 ± 0.9	0.25
HOMA-IR	3.4 ± 1.2	3.5 ± 1.4	0.56
AST (IU/l)	21 ± 6.5	23 ± 10.2	0.43
ALT (IU/l)	23 ± 14.8	25 ± 16.8	0.38
Total cholesterol (mmol/l)	4.5 ± 0.9	4.6 ± 0.8	0.49
Triglyceride (mmol/l)	1.5 [1.1, 2.5]	1.5 [1.1, 2.1]	0.35†
HDL-cholesterol (mmol/l)	1.2 ± 0.3	1.3 ± 0.3	0.47
LDL-cholesterol (mmol/l)	2.4 ± 0.7	2.6 ± 0.7	0.12
Adiponectin (μg/ml)	3.1 [1.7, 4.6]	2.9 [1.6, 4.6]	0.71†
hsCRP (μg/ml)	1.0 [0.5, 1.6]	0.8 [0.4, 1.8]	0.37†
Interleukin-6 (pg/ml)	1.4 [0.9, 2.1]	1.1 [0.8, 1.7]	0.12†

Data are presented as mean ± SD, median [1st quartile, 3rd quartile], or number of subjects (%). All *P*-values were obtained from independent two-sample *t*-tests except as indicated: *Chi-squared test and †Mann–Whitney *U*-test.
BMI, body mass index; baPWV, brachial-ankle pulse wave velocity; HOMA-IR, homeostasis model assessment insulin resistance; AST, aspartate aminotransferase; ALT, alanine aminotransferase; HDL, high-density lipoprotein; LDL, low-density lipoprotein; hsCRP, high-sensitivity C-reactive protein.

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Table 2 Changes of anthropometric and laboratory parameters by group after 3 months

	Control group (n = 54)			Intervention group (n = 57)		
	Baseline	After 3 months	P-value	Baseline	After 3 months	P-value
Weight (kg)	67.7 ± 10.8	66.4 ± 10.4	0.004	66.4 ± 12.5	65.3 ± 12.7	0.002
BMI (kg/m^2)	25.5 ± 3.3	25 ± 3.3	0.001	25.6 ± 3.5	25.1 ± 3.5	0.001
Waist circumference (cm)	91.3 ± 7.5	89.1 ± 7.6	0.001	89.5 ± 9.7	86.8 ± 9.8	< 0.001
Systolic blood pressure (mmHg)	1387.8 ± 17.8	134 ± 13.64	0.115	140.2 ± 187.6	1332.7 ± 16.2	0.001
Diastolic blood pressure (mmHg)	832.9 ± 10.0	82.2 ± 7.7	0.648	843.8 ± 10.0	80.3 ± 9.2	0.005
Right baPWV (cm/sec)	1691 ± 377	1596 ± 318	0.002	1671 ± 382	1564 ± 271	< 0.001
Left baPWV (cm/sec)	1695 ± 386	1608 ± 333	0.002	1648 ± 335	1569 ± 260	0.001
HbA_{1c} (%)	7.4 ± 0.9	7.6 ± 1.0	0.033	7.6 ± 0.9	7.1 ± 0.8	< 0.001
Fasting glucose (mmol/l)	7.5 ± 1.8	7.5 ± 1.9	0.851	7.9 ± 1.8	7.5 ± 1.6	0.156
HOMA-IR	3.4 ± 1.4	3.4 ± 1.2	0.877	3.5 ± 1.4	3.6 ± 1.5	0.906
Total cholesterol (mmol/l)	4.5 ± 0.9	4.5 ± 0.8	0.882	4.6 ± 0.8	4.1 ± 0.7	< 0.001
HDL-cholesterol (mmol/l)	1.2 ± 0.3	1.3 ± 0.3	0.010	1.2 ± 0.3	1.3 ± 0.3	0.001
LDL-cholesterol (mmol/l)	2.4 ± 0.7	2.3 ± 0.7	0.628	2.6 ± 0.7	2.2 ± 0.6	< 0.001
Triglyceride (mmol/l)	1.51 [1.1, 2.5]	1.59 [1.2, 2.4]	0.896	1.46 [1.1,2.1]	1.24 [0.8,1.8]	0.003#
Adiponectin (μg/ml)	3.1 [1.7, 4.6]	3.5 [2.0, 7.0]	0.267	2.9 [1.6, 4.6]	7.5 [4.6, 14.5]	< 0.001#
hsCRP (μg/ml)	1.0 [0.5, 1.6]	0.7 [0.3, 1.4]	0.099	0.8 [0.4, 1.8]	0.5 [0.2, 1.3]	0.090#
Interleukin-6 (pg/ml)	1.4[0.9, 2.1]	1.4 [0.9, 1.9]	0.918	1.1 [0.8, 1.7]	1.0 [0.6, 1.7]	0.196#

Data are presented as mean ± SD or median [1st quartile, 3rd quartile]. P-values for changes after 3 months within each group were calculated using a paired sample t-test or Wilcoxon's signed rank test (#).
BMI, body mass index; baPWV, brachial-ankle pulse wave velocity; HOMA-IR, homeostasis model assessment insulin resistance; HDL, high-density lipoprotein; LDL, low-density lipoprotein; hsCRP, high-sensitivity C-reactive protein.

group during the last week was also significantly lower than in the first week. Systolic blood pressure dropped significantly from 127 ± 14 mmHg to 120 ± 19 mmHg (P = 0.001). Diastolic blood pressure decreased from 78 ± 10 mmHg to 74 ± 8 mmHg (P < 0.001). In addition, total cholesterol, LDL-cholesterol and triglyceride levels in the intervention group were significantly decreased after 3 months (Table 2). There were no significant changes in these factors in the control group.

HbA_{1c} in the control group rose significantly after 12 weeks (7.4 ± 0.9% to 7.6 ± 1.0%, P = 0.03). In contrast, HbA_{1c} improved in the intervention group (7.6 ± 0.9% to 7.1 ± 0.8%, P < 0.001). Moreover, in the intervention group, the average self-monitoring blood glucose (SMBG) level during the last week was significantly lower than the average during the first week. Fasting SMBG levels were significantly decreased from 7.4 ± 1.5 to 6.8 ± 1.5 mmol/l (P = 0.01). SMBG levels before bedtime were also significantly reduced from 9.0 ± 1.8 to 8.1 ± 1.8 mmol/l (P = 0.001).

HbA_{1c}, total cholesterol and LDL-cholesterol levels were significantly decreased after 3 months in the intervention group compared with the control group on repeated measures ANOVA method (Fig. 3).

Changes in adiponectin, proinflammatory markers and arterial stiffness

In the intervention group, but not the control group, there were significant increases in adiponectin levels after implementation of the UCDC system (Table 2; Fig. 3). In both the control and intervention groups, baPWV improved significantly, although hsCRP and IL-6 levels did not change in either group (Table 2).

Safety and compliance evaluation

Five patients (8.1%) dropped out of the intervention group and seven (10%) out of the control group. The characteristics of patients who did and did not drop out were similar in both the intervention and control groups. Participants in the intervention group sent blood glucose recordings 1.84 ± 0.31 times per day (compliance rate 92.2 ± 15.4%) and blood pressure 1.72 ± 0.32 times per day (compliance rate 86.0 ± 16.2%). Body weight measurements were sent 0.87 ± 0.20 times per day (compliance rate 87.4 ± 20.1%). During the trial, drug modification and significant side-effects, including hypoglycaemia, were not reported in either group.

Discussion

The results of the study show that a novel chronic disease care system using both cellular phones and the internet contributes to the improvement of multiple metabolic factors in patients with multiple chronic diseases in both community-based and university hospital settings.

Although the incidence and risk of chronic disease continues to increase dramatically, current efforts to control these diseases and prevent cardiovascular events have been disappointing. According to NHANES 1999–2000, only 7.3% of adults with diabetes simultaneously achieved the recommended goals of HbA_{1c} > 7.0%, blood pressure

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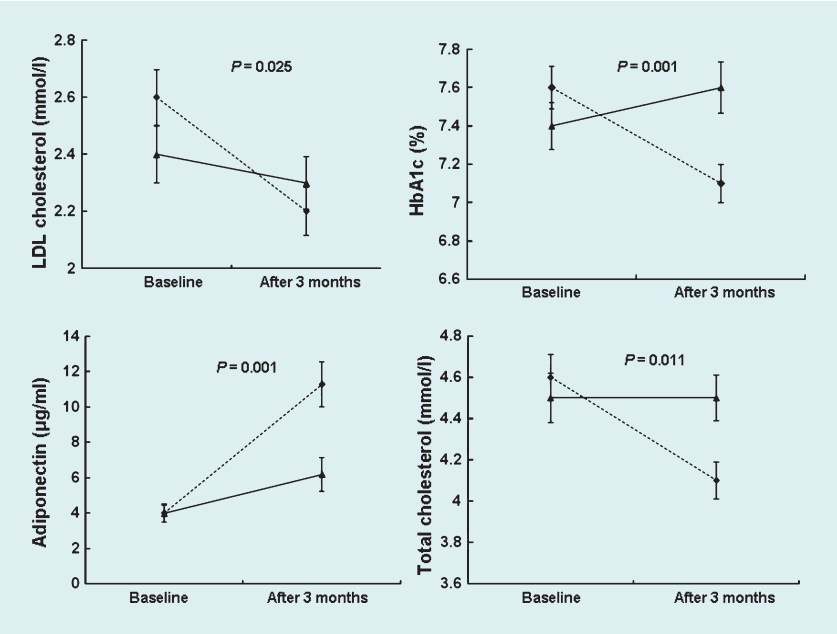

FIGURE 3 Changes in various parameters between baseline and 3 months in the intervention (dashed lines) and control groups (solid lines). *P*-values for the interaction effect between time and group were obtained using the repeated measures ANOVA method after adjusting for age, gender, current smoker status, the duration of diabetes, and change of weight. The original adiponectin measurements were transformed to the ranked ones before employing the repeated measures ANOVA method.

> 130/80 mmHg, and total cholesterol > 5.8 mmol/l [19]. This finding demonstrates that the current healthcare system is limited in its capacity to facilitate adequately self-management of these chronic diseases. To manage future costs better and improve the quality of care, healthcare must shift from the individual, acute care model to a community-based, chronic and multiple disease care system.

Most chronic diseases, including diabetes, hypertension and obesity, often require lifestyle intervention supported by frequent feedback from care providers. Motivated patients who self-monitor frequently, and have access to ongoing advice from a health professional, may be able to achieve the appropriate behaviour changes and better glycaemic control, as observed in the Diabetes Control and Complication Trial and the Stockholm Diabetes Intervention Study [20,21]. Web or cellular phone-based intervention is an efficacious method for continuing the educational, motivational and monitoring activities of patients with chronic illnesses. Cho *et al.* [4] demonstrated that long-term use of an internet-based glucose monitoring system was superior to a conventional diabetes care system for controlling blood glucose and achieving glucose stability. Moreover, Harvey-Berino *et al.* [22] reported that participants assigned to an internet-based weight maintenance programme sustained weight loss over 18 months comparable to individuals who continued to meet their doctor face-to-face. Cellular phone-based intervention has generally also been positive, although some negative studies have been reported. Logan *et al.* [13] recently reported that a mobile phone-based remote blood pressure-monitoring system lowered blood pressure in diabetic patients, and Montori *et al.* [14] showed that poorly controlled Type 1 diabetic patients receiving telecare had a significantly lower HbA_{1c} after 6 months compared with the usual care group.

However, there are many barriers to introducing this chronic care system to the community healthcare system. One potential barrier is the high attrition rate caused by frustrations related to the input of data after logging on to the website [23]. Attrition rates as high as 34% have been reported in weight loss trials when using the internet for the input of data [24]. Another potential barrier to the use of e-medicine is the lack of structured and immediate feedback [25]. Increasing the amount of personalized feedback could be key to increasing compliance and the success of telemedicine-based chronic disease control [26].

With these concerns in mind, our UCDC system used cellular phones for the input of metabolic parameters. Patients also received immediate and structured feedback from their physicians using text messages. Frequent self-monitoring, and immediate medical advice according to their most recent data, may have motivated the patients to control their chronic diseases more enthusiastically.

The patients receiving care through the novel chronic disease care system showed significant improvements in four major categories compared with those in the control group. HbA_{1c}, total cholesterol, LDL-cholesterol and adiponectin levels were all improved using this new care system. LDL-cholesterol accounts for most of the lipid risk in predicting early cardiovascular disease [27]. Adiponectin is the major anti-inflammatory and anti-atherogenic adipokine that is secreted by adipose tissue [28]. Previous studies have

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reported that serum adiponectin levels may predict future cardiovascular mortality and morbidity [29]. Furthermore, the UCDC system improved both HBPM and OBPM. HBPM is less susceptible to observer bias and the white coat effect than OBPM. HBPM also correlates better with end organ damage [30].

In the present study, although the participants were relatively old, all but five subjects in the intervention group completed the study with high compliance. Overall, participants in the intervention group did not find the UCDC system difficult to use. They did not experience significant side-effects, and were satisfied with the continuous care of their chronic disease. Participants in our study had a cellular phone, allowing them to input their data conveniently using text messages. This also gave them the ability to receive immediate recommendations by text messages based on the data they provided. This easy-to-use interface could make patients more likely to be comfortable with and utilize an e-medicine system. Our UCDC system also improved the utilization of medical resources. Our system used an automatic algorithm for notification, confirmation and feedback recommendations based on the measurement of metabolic parameters.

There were some limitations to our study. First, the cost–benefit ratio of using the UCDC system in the management of chronic disease was not calculated. A study on the economic feasibility of using this healthcare delivery system is needed to determine the qualitative merits of electronic medicine. Second, the metabolic benefits of the UCDC system were examined only over a 3-month interval. It is not clear if these effects would persist over a longer period of time.

In conclusion, a UCDC system using cellular phones and the internet improved multiple metabolic parameters and had beneficial effects on several cardiovascular risk factors in overweight patients with both Type 2 diabetes and hypertension over a 3-month period. This system may improve multifactorial care in patients with multiple chronic diseases. Long-term, large-scale studies and qualitative investigations are needed to determine further the utility of this novel chronic disease care system.

Competing interests

Nothing to declare.

Acknowledgements

This study was funded by a grant from the Seoul R & BD Project. The development of the HSA business model and technology was sponsored by the Ministry of Commerce, Industry and Energy. The funding sources had no role in the study design. They had no control over the collection, analyses, or interpretation of all study data, nor did they control the writing of the article or the decision to submit the manuscript for publication.

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